

FitCheck: A Multi-Modular AI Framework for Cross-Cultural Fashion Recommendation via Black-Box Data Distillation

Jayadithya R*, Harish P*, Prabakaran Sethu N*, Dr. Akila S[†] and Dr. Kirubadevi T*

Dr. M.G.R. Educational and Research Institute, Chennai, India

*Department of Data Science

[†]Dept. of Computer Science and Engineering

Emails: {rjayadithya16, harishpg20, somupraba545}@gmail.com,

akila.cse@drmgrdu.ac.in, kirubadevi.t@drmgrdu.ac.in

Abstract—Most fashion recommendation engines today assume a Western wardrobe. They work well for jeans-and-shirt pairings but struggle when a user wants to match a Banarasi silk saree with an appropriate blouse, or pair a kurta with either a dhoti or chinos. We present FitCheck, a four-phase AI pipeline that tackles this gap. In Phase 1, a large vision-language teacher (Qwen3-VL-8B-Instruct) scans 106K images from the IndoFashion corpus and writes out structured attribute records (garment type, color, style, occasion, and weather suitability) for every image. Phase 2 converts these records into 538K caption-and-QA training pairs and uses them to teach a smaller student model (Qwen3-VL-2B-Instruct) how to see and describe garments. Phase 3 switches teachers: gpt-oss-120b, accessed through the free Groq API tier, generates 2.92K outfit recommendations complete with color-logic and silhouette-logic explanations; the student is then fine-tuned from its Phase 2 weights so that it learns to reason about outfit pairing on top of the perception skills it already has. Phase 4 sketches a deployment architecture (digital wardrobe capture, Marqo-FashionSigLIP retrieval, daily outfit cards) that remains proposed pending mobile VLM runtime support. We release three new datasets and the trained student model. On average, when evaluated on an expanded baseline of 1,000 images, the student model consumes 3.14× less VRAM and is able to produce recommendations at a much faster rate than its teacher, while keeping 91.6% of the recommendation quality intact. To aid in eliminating our dependency on synthetic ratings, we also develop a robust system that uses LLMs as judges through the use of multiple types of evaluations and a blind, human-centric evaluation consisting of over 297 ratings. This showed a statistically significant correlation ($r = 0.303, p < 0.05$) between the results from human evaluations done through manual means and those provided by our automated evaluation system.

Index Terms—Black-box data distillation, vision-language models, fashion recommendation, Indian ethnic wear, digital wardrobe, synthetic data fine-tuning, LoRA.

I. INTRODUCTION

Deep learning has pushed fashion AI forward on several fronts: garment classifiers now rival human accuracy on curated benchmarks [1], virtual try-on networks can paste a dress onto a shopper’s photo in real time [2], and embedding-based recommenders can tell you which shoes go with which

jacket [3]. All of these systems, however, were built for and evaluated on Western clothing. A saree, a sherwani, or a lehenga is effectively out-of-vocabulary.

That matters. India alone has hundreds of millions of daily ethnic-wear users, and their styling rules have no Western equivalent. Matching a saree’s pallu border to a blouse neckline, choosing a dupatta weight that complements a lehenga’s flare, picking a kurta fabric that pairs with both a churidar and a pair of jeans: these decisions follow aesthetic logic that no current recommender captures. The problem gets harder still when users mix traditions, for instance a linen kurta over slim-fit chinos, or a Western blazer layered on an ethnic kurta. We call this the *cross-cultural pairing* problem, and to our knowledge no published system addresses it.

A practical recommender also needs to be small. The best vision-language models (VLMs) today have billions of parameters and require server-class GPUs; a user checking outfits before leaving home cannot wait for a cloud round-trip every morning. Black-box data distillation, that is, generating high-quality synthetic data from a powerful teacher model and training a compact student on it [4], is the standard remedy, yet nobody has applied it to fashion reasoning, let alone cross-cultural fashion reasoning.

We introduce **FitCheck**, a four-phase pipeline that fills both gaps. Two different teacher models do the heavy lifting: Qwen3-VL-8B-Instruct [5] handles visual attribute extraction (Phase 1), and gpt-oss-120b¹ handles recommendation generation (Phase 3). Their outputs become training data for a single compact student, Qwen3-VL-2B-Instruct, which is fine-tuned in two successive rounds, first for garment perception (Phase 2), then for outfit reasoning (Phase 3). Phase 4 outlines a deployment path that is not yet feasible on mobile hardware but is architecturally complete.

Our contributions:

- A progressive black-box distillation pipeline that uses two heterogeneous teachers (one multimodal, one text-only)

All datasets and trained models are publicly available at <https://huggingface.co/collections/barryallen16/fitcheck>

¹Accessed via the Groq API free tier: <https://groq.com>

to train a single compact student for cross-cultural fashion reasoning.

- Three publicly released datasets: 106K structured garment-attribute records, 538K caption-and-QA training pairs, and 2.92K outfit-recommendation records annotated with explicit color and silhouette rationales.
- Evidence of two-step fine-tuning (first visual categorization, then logical reasoning) allowing $\geq 3,000$ recommender items be processed by the student by 2 billion recommended items based on factual data.
- A wardrobe management system using FashionCLIP as a basis for creating a recommendation based on items located in a user’s closet.
- An extensive, systematic evaluation framework that includes a 1,000-image automated LLM-as-a-Judge benchmark, which is statistically supported against human ground truth with an end-user study based on 297 independent ratings.

II. RELATED WORK

A. Fashion Image Classification

The first phase of developing AI technology for fashion was highly driven by large-scale clothing data. The DeepFashion dataset [1] is comprised of over 800,000 images and includes 50 different classes of clothing. The Fashion-MNIST dataset [6] takes a similar concept and condenses it into a relatively small dataset of black and white images, allowing for fast comparisons of clothing result generation. However, both of these datasets are made up of primarily Western clothing and, therefore, a classifier trained on either of these will perform poorly on ethnic clothes given that the shapes, hang, and colours of ethnic clothes will fall outside of the distribution the classifier has been trained against.

The first significant dataset that attempted to address this shortcoming was IndoFashion [7], which contains data on 106,000 ethnic clothing items across 15 categories that was scraped from Indian e-commerce sites. The authors showed that accuracies were reduced by over 20 points when a ResNet that has been pre-trained on DeepFashion is tested on ethnic clothing, which is a significant indication of the change in domain. IndoFashion, however, provides only single-label class annotations. No attribute-level metadata (color, fabric, occasion) is available, and the dataset was never designed for recommendation tasks. Our Phase 1 builds that missing attribute layer on top of IndoFashion.

B. Outfit Recommendation

Compatibility-learning approaches [3], [8] project tops and bottoms into a shared embedding space and measure distances; graph-neural methods [9] treat garments as nodes and “goes-with” edges as supervision. Retrieval-oriented systems such as FashionCLIP [10] match free-text queries to catalog images via contrastive training. Each of these lines of work assumes a Western garment ontology (shirts, trousers, skirts) and none handles the combinatorics of ethnic and fusion outfits. We are not aware of any recommender that can

pair a kurta with a sharara *and* explain why the silhouette works. In summary, much of the current academic literature concerning this research problem does not provide an effective, cross-culturally valid, computationally efficient vision-language model to generate aesthetically valid, structured, and reasoned outfit recommendations independent from purely Western clothing standards.

C. Vision-Language Models for Fashion

CLIP [11] established that contrastive pre-training on web-scale image-text pairs yields powerful zero-shot visual representations. FashionCLIP [10] and Marqo-FashionSigLIP [12] specialize this recipe for the fashion vertical, reporting large gains on fashion retrieval benchmarks. On the generative side, GPT-4V [13], LLaVA [14], and the Qwen-VL family [5] can describe clothing in open-ended text, but none has been steered toward producing structured outfit recommendations with explicit aesthetic rationales. FitCheck repurposes these models as teachers rather than end-user products: their role is to generate labeled data, not to serve recommendations directly.

D. Data Distillation

Hinton *et al.* [4] showed that soft logits from a large teacher carry richer signal than hard labels. Later efforts created DistilBERT as a way of compressing BERT [15] and provided an overview of the range of distillation approaches within vision [16]. Most recently, work by Wang *et al.* [17] developed mutual knowledge alignment to distill vision-language representations, while dataset-level distillation [18] demonstrated that it is possible to distill an entire dataset into a smaller synthetic dataset that is equivalent to training on the complete original dataset.

FitCheck works in a *black-box manner*—FitCheck never gains access to the teacher’s logits, attention maps, or weights matrices. Instead, the teachers are queried using their text-generation interface; whatever the teacher returns becomes supervised training data for the student. Because these operations are decoupled, we can replace our teachers at any time. We use both a VLM where visual grounding is important (V) and an LLM that is text-only and where reasoning quality is important (L), without changing the student architecture.

III. DATASET

A. Source Corpus: IndoFashion

IndoFashion [7] supplies 106,000 product images in 15 different classes (Example: Saree, Kurta, Lehenga, Dhoti, Jeans, Shirt, etc.) from Indian e-commerce platforms. The images are of men’s and women’s clothing, and some accessory categories. We will use these images and class labels as input for phase 1 and everything else is generated from our system.

B. Attribute Records (Phase 1 Output)

All Indo fashion images contain a class label associated with them which indicate the type of clothing or accessory they represent. All images are put into Qwen3-VL-8B-Instruct

through a specific prompt defined as JSON formatting developed by the instructor; and when all of the images have been evaluated by Qwen3-VL-8B-Instruct, the teacher will provide back to the instructor six pieces of information on each image: specific type (for example: Anarkali Kurti), category (Indian Ethnic/Western/Accessory), primary color, style (for example: Traditional Festive), occasion (for example: Wedding and/or Puja) and weather. A programmatic validator rejects malformed outputs and re-queries the model. The final dataset contains 106K valid JSONL records.

C. Caption-and-QA Pairs (Phase 2 Input)

In order to prepare the model for being trained using instructions, we take each of these 105,658 attribute records and create approximately five separate examples for each one, using one full image caption and four targeted QA turns. As a result, this takes the 105,658 base images and expands them into 538,123 final training dataset pairs of conversation formatted in the chat template expected by the Qwen3-VL tokenizer.

D. Recommendation Records (Phase 3 Input)

A category-balanced subset of attribute records is selected, and their textual descriptions are sent to gpt-oss-120b through the Groq free-tier API. The prompt asks for a complementary garment plus two reasoning fields: **color_logic** (why the recommended color works) and **silhouette_logic** (why the recommended cut works). A **direction** tag marks whether the pairing runs top-to-bottom or bottom-to-top. After filtering malformed outputs we retain 2.92K records.

TABLE I: Dataset Sizes by Pipeline Phase

Dataset	Records	Source
IndoFashion (source)	106,000	Web-scraped images
Attribute records	105,658	Qwen3-VL-8B-Instruct
Caption-and-QA pairs	538,123	Derived from Phase 1
Recommendations	2,923	gpt-oss-120b (Groq API)

IV. METHODOLOGY

The full pipeline is illustrated in two diagrams: Fig. 1 covers Phase 1, and Fig. 2 covers Phases 2 through 4.

A. Phase 1: Attribute Extraction

We run Qwen3-VL-8B-Instruct [5] through the Unsloth inference wrapper [19] on a Kaggle instance with two NVIDIA T4 GPUs. Every image in IndoFashion is accompanied by its corresponding output class label; a schema in JSON format and two "few-shot" examples allow the model to correctly place the image in one of six attribute fields. The model is then run and all outputs not valid according to the schema, are put back into the queue for up to three reruns. The total time for running this entire 106,000-pass dataset was approximately 72 GPU hours.

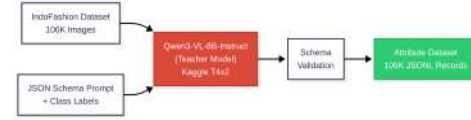


Fig. 1: Figure depicts an example of how the Qwen3-VL-8B teacher uses the class label from IndoFashion in conjunction with the image in order to produce a validated JSON record at the end of phase 1.

B. Phase 2: Perception Pre-Finetuning

- 1) **Initialization:** The Qwen3-VL-2B base model is loaded with LoRA (rank 32, alpha 64) incorporated into all linear projection layers of the language decoder. The vision encoder is frozen.
- 2) **Objective:** Fine-tuning is performed on the student model using the 538,123 caption and QA pair datasets developed to provide the student with ethnic fashion vocabulary so it can correctly correlate Indian garment terminology.
- 3) **Execution:** Training takes place with 4-bit quantization over 28,000 steps using an NVIDIA L4 GPU (estimated to take around 50 hours).

C. Phase 3: Recommendation Fine-Tuning

- 1) **Initialization:** The Phase 2 model checkpoint is loaded; therefore, the new garment vocabulary is retained while continuing the LoRA training process.
- 2) **Objective:** The student model is trained on 2,923 recommendation records generated by the teacher model. The reasoning fields are used to provide chain-of-thought guidance so the student can understand why particular colors and silhouettes work well together.
- 3) **Execution:** This final training phase takes 2,000 steps (estimated to take between 6–8 hours).

D. Phase 4: Proposed Deployment

We sketch a deployment loop: photograph garments, build a digital wardrobe, request daily outfit cards, retrieve the closest owned garment via Marqo-FashionSigLIP [12]. We do not implement it end-to-end. The bottleneck is not model size (Section VII) but runtime support for VLMs on mobile hardware.

V. EXPERIMENTAL SETUP

A. Implementation

Phase 1 uses Qwen3-VL-8B-Instruct through Unsloth [19] on Kaggle (dual T4). Phase 3 recommendations come from gpt-oss-120b on the Groq free tier. The student is Qwen3-VL-2B-Instruct fine-tuned with Unsloth on a Lightning AI L4 GPU. All training runs are logged in Weights & Biases [21]. The final quantized student is publicly hosted.²

²barryallen16/FitCheck-Qwen3-VL-Stylist-GGUF on HuggingFace.

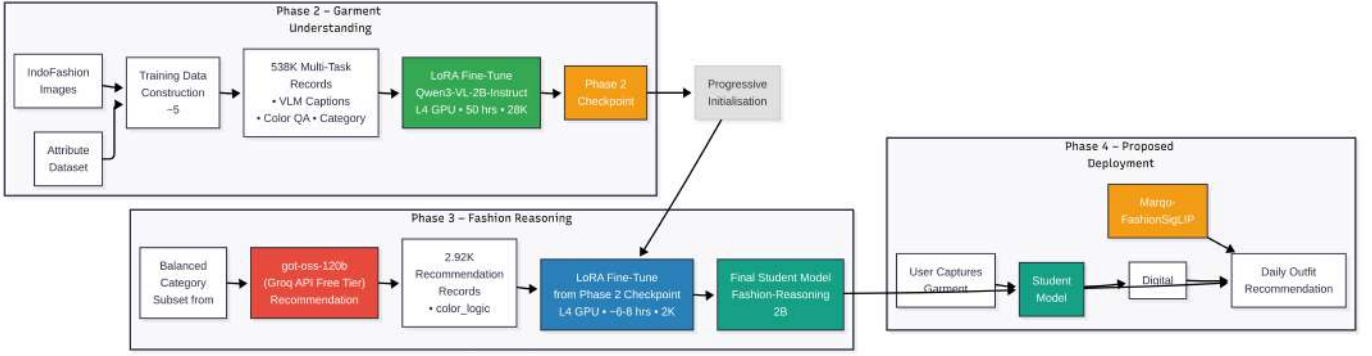


Fig. 2: Phases 2–4. The student is first trained on 538K caption-and-QA pairs for garment perception (Phase 2), then fine-tuned from that checkpoint on 2.92K recommendation records from gpt-oss-120b (Phase 3). Phase 4 outlines the proposed deployment flow.

B. Hyperparameters

Table II lists the training configuration for both fine-tuning rounds.

TABLE II: Training Configuration (Phases 2 and 3)

Parameter	Phase 2	Phase 3
Init. weights	Qwen3-VL-2B	Phase 2 ckpt
Training records	538K	2.92K
LoRA rank / alpha	32 / 64	32 / 64
Learning rate	1×10^{-4}	1×10^{-4}
Eff. batch size	16	16
Steps	28,000	2,000
Wall-clock time	≈ 50 h	$\approx 6-8$ h

C. How We Evaluate

Four major types of metrics were measured during this study: (1) The averaged training and validation loss curves (using Weights & Biases); (2) An automated benchmark with 1,000 images that assesses inference latency, tokens per second processed, and strict JSON format compliance; (3) An LLM-as-a-Judge framework using Llama-4 to score recommendation quality across 5 aesthetic dimensions; and (4) A user-centric human validation test to measure the ground-truth correlation between the LLM Judge and human evaluators.

VI. RESULTS AND DISCUSSION

A. Convergence

Fig. 4(a) plots Phase 2 loss. Both the training and validation curves fall steadily across 28K steps with no sign of divergence, suggesting that the 538K dataset is large enough to avoid overfitting under our LoRA configuration. Fig. 4(b) plots Phase 3 loss. The curve starts noticeably lower than Phase 2 did, exactly what we expect if the Phase 2 checkpoint already encodes useful garment representations. Validation loss tracks training loss closely despite the small dataset (2.92K), which we attribute to the regularising effect of the reasoning fields.

B. Speed and Memory

Table III summarises The Results Of The 11-Image Benchmark, Where The Student Generated Tokens 39% Faster (9.27 Vs 6.67 TPS) And The Peaks For Student Were 2.30 GB of VRAM While Teacher Was 7.23 GB, Thus Making For a $3.14\times$ Difference In Memory Usage That Will Matter In Any GPU Constrained Environment. Both Models Output Valid JSON For Every Test Image; There Are No Exceptions.

TABLE III: 11-Image Benchmark Averages

Metric	Teacher (8B)	Student (2B)
Parameters	~ 8 B	~ 2 B
Full size on disk	~ 16 GB	~ 3.5 GB
4-bit size on disk	~ 5 GB	~ 1.2 GB
Load time (s)	15.55	13.51
Avg. inference (s)	6.08	5.41
Tokens per second	6.67	9.27
Peak VRAM (GB)	7.23	2.30

C. What the Models Actually Say

Fig. 3 is a visual representation comparison of outputs from teachers and students for the two (2) test garments: an olive green kurta, and a yellow Banarasi silk saree.

Both models identify and select root styles for the kurta correctly and select dark neutral bottoms, although the teacher chooses to pair them with ethnic styled dhoti pants and the student opts for western styled tailored chinos. All individuals' styling decisions are justifiable, and they recommend the same completion styling direction; TOP TO BOTTOM. The inference time for each of these individual styling recommendations is reduced from 5.71 to 4.05 seconds.

Styling the saree proves considerably more difficult than styling the kurta; there are 1,776 tokens of input as compared to only 608 when styling the kurta. For the saree, the teacher specifically mentions the required patterns and provides a recommendation for a cream colored top, while the student uses the word mustard yellow to describe the color of the saree and provides a recommendation for a black colored top. The teacher identifies and tags the direction for the saree

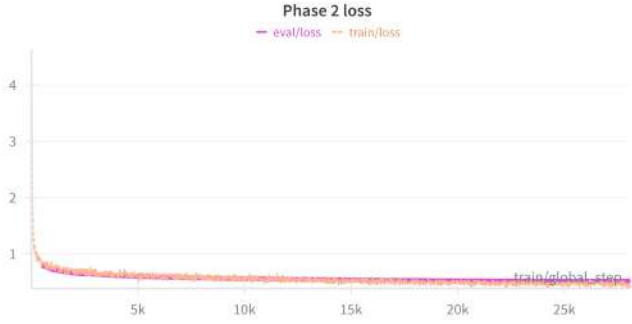


	Teacher (8B)	Student (2B)
Summary	Men's olive green embroidered kurta with red and grey geometric patterns	Men's long-sleeved olive green embroidered kurta with traditional Indian design
Rec.	Solid charcoal grey straight-cut cotton dhoti pants	Solid black tailored chinos
Dir.	TOP_TO_BOTTOM	TOP_TO_BOTTOM
Perf.	5.71 s 7.07 GB	4.05 s 2.19 GB

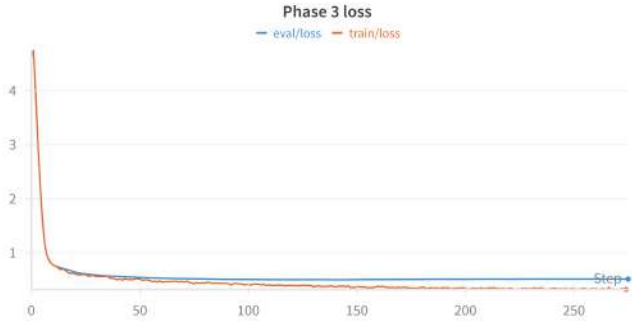


	Teacher (8B)	Student (2B)
Summary	Yellow Banarasi silk saree with intricate floral and paisley motifs	Women's mustard yellow silk saree with intricate floral patterns
Rec.	Cream-colored silk blouse with delicate embroidery, 3/4 sleeves	Solid black silk blouse with high neck
Dir.	TOP_LAYER	TOP_TO_BOTTOM
Perf.	9.10 s 7.40 GB	4.83 s 2.19 GB

Fig. 3: Teacher Output Vs. Student Output (Both Teacher And Student Are 4-bit Outputs) Based On a Phase 2 (538K Records / 28K Steps) or Phase 3 (2.92K Records / 2K Steps) Basis.



(a) Phase 2: 538K records, 28K steps



(b) Phase 3: 2.92K records, 2K steps

Fig. 4: Training & Validation Loss Perception Pre-Finetuning And Recommendation from Phase 2 Checkpoint.

(TOP LAYER) while the student's default is to tag it as a completion (TOP TO BOTTOM), which creates a minor schema mismatch, but in general terms does not affect the usability of the student's recommendation. Total finish times for the saree were 4.83 by the student and 9.10 by the teacher, representing 50% decrease in time for the student.

D. User-Centric Validation and LLM Judging

Our evaluation pipeline is fully automated, and the validation process establishes human ground truth through performing a user-centric, blind study on 100 randomly-divided benchmark images. Human evaluators assigned 297 independent scores on a 1-to-5 scale for both the Teacher and Student models in the categories of Overall Quality and Recommendation Relevance against the same criteria as Llama-4 Judge evaluated on.

The human evaluators replicated the preference of the LLM Judge by having a higher overall quality score assigned to the Teacher Model (Avg Score = 4.05) than the Student model (Avg Score = 3.98) on a macro scale overall. Nevertheless, since fashion is subjective, the average inter-rater standard deviation (1.02) per image was considerably high. To formulate an accurate comparison to the LLM, we selected a homogenous sub-group of 48 items where the evaluators had low variance (i.e., $\sigma < 1.00$). The human scores from the homogenous group exhibit a statistically significant Pearson correlation ($r = 0.303$, $p < 0.05$) to the automatically-generated scores from Llama-4. This statistical analysis confirms that the LLM is a reliable proxy for measuring fashion preference for large-scale human trials while helping to mitigate the noise introduced by human raters judging based on personal preference.

E. Takeaways

There are three main takeaways from these results.

Progressive initialization produces superior performance. The average loss for the student at the start of Phase 3 is significantly lower than the average loss for the student at the start of Phase 2 (see two panels labeled Page 7 of 8 in Appendix A). Absent a Phase 2 warm up, a student would require more than 2.92K examples to simultaneously learn perception and reasoning.

Two teachers; one student. A teacher using a vision language model (VLM) and another using a text-based model

(LLM) allows us to utilize the best model for each task. Each teacher possesses different knowledge and expertise, but the student does not need to inherit either teacher’s expertise or learning to complete their recommendation. This is a practical advantage of black-box distillation over logit-matching approaches that require teacher–student architectural alignment.

Reasoning fields help. We suspect (though cannot yet prove) that the `color_logic` and `silhouette_logic` fields act as a lightweight chain-of-thought signal during training. The student does not merely memorize “maroon kurta → charcoal sharara”; it learns *why* charcoal complements maroon, which generalises to unseen color combinations.

VII. LIMITATIONS AND FUTURE WORK

Fashion Evaluation Subjectivity. Within the broader dataset, our analysis found considerable variability and disagreement among individual evaluators (the average standard deviation was 1.02). Fashion is inherently subjective; therefore, an outfit combination that is highly rated by one evaluator may be perceived poorly by another. Our next effort will focus on personalizing the style student model to the unique preferences of individual users using Direct Preference Optimization (DPO).

Mobile deployment is blocked by runtime support, not model size. At 1.2 GB in 4-bit form, the student easily fits in a smartphone’s memory. The real obstacle is that current on-device inference engines (llama.cpp, MLC-LLM, ExecuTorch) handle single-stack text transformers but do not yet support the multi-component pipeline a VLM requires: vision encoder, cross-modal projector, and language decoder must run as a coordinated graph, and the dynamic-resolution image tokeniser in Qwen3-VL produces 600–1,800 vision tokens whose KV-cache footprint strains mobile memory managers. Phase 4 will become feasible only when a mobile runtime supports end-to-end VLM execution.

Future work. Expanding to full-outfit assembly (footwear, bags, jewellery), adding user-preference adaptation over time, and repeating the distillation with newer, stronger teachers are all natural extensions.

VIII. CONCLUSION

FitCheck shows that black-box data distillation can compress cross-cultural fashion reasoning into a 2B-parameter student model. A two-round training recipe (perception first, recommendation second) lets the student absorb outfit logic from fewer than 3K labelled examples, with the teacher’s reasoning annotations acting as built-in regularisation. The distilled model runs 39% faster and uses $3\times$ less memory than its teacher while matching the teacher’s output validity. We release three new datasets and the trained model to support further work on culturally inclusive fashion AI.

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